1.08 Graph Sketching: Quadratic Graphs

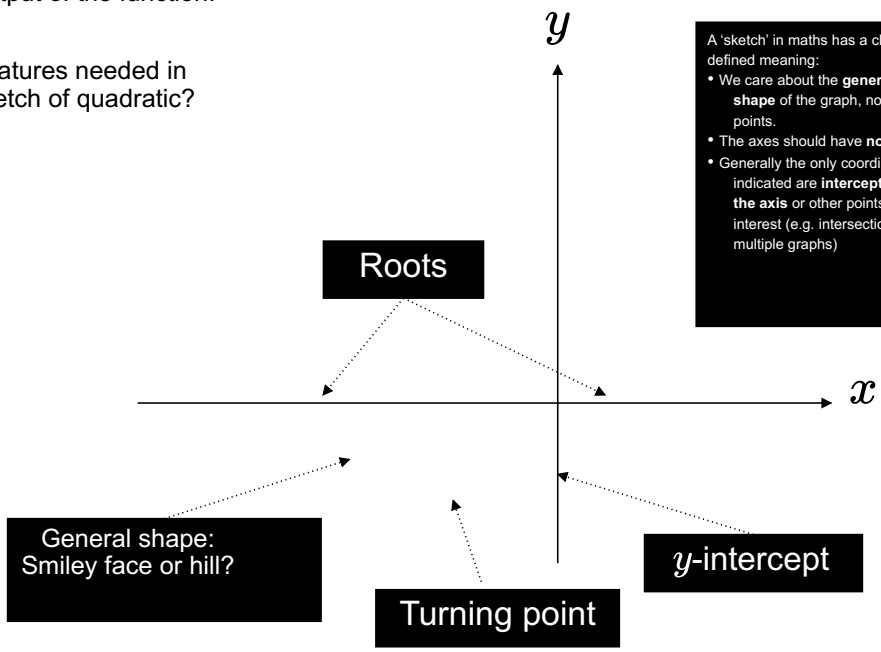
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4: Quadratic Graphs

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Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

Features needed in sketch of quadratic?



A 'sketch' in maths has a clearly defined meaning:
- We care about the **general shape** of the graph, not exact points.
- The axes should have **no scale**.
- Generally the only coordinates indicated are **intercepts with the axis** or other points of interest (e.g. intersections of multiple graphs)

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Example

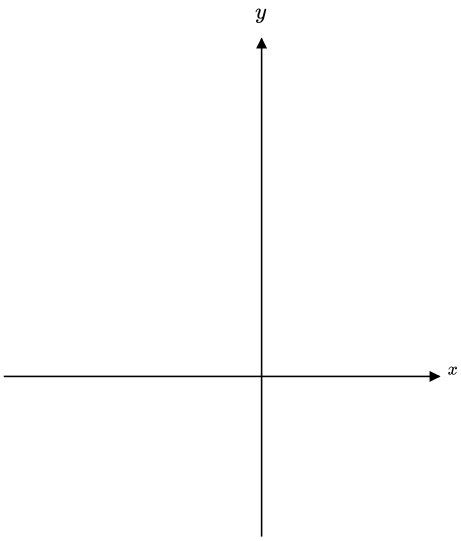
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Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

y-intercept:

Turning point:



Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

1.08 Graph Sketching: Quadratic Graphs

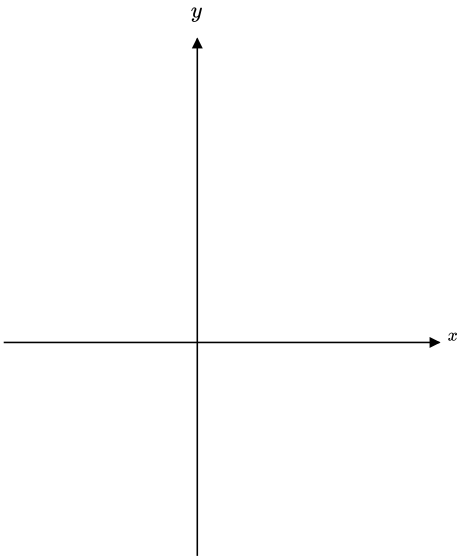
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Example

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Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

Roots:



y-intercept:

Turning point:

Line of reflection:

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Test Your Understanding

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Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

a $y = x^2 + 4$

c $y = 5x + 3 - 2x^2$

b $y = x^2 - 7x + 10$

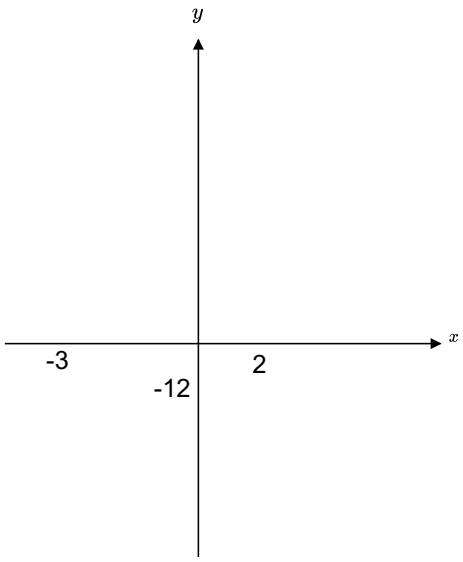
d $y = x^2 + 4x + 11$

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Determining the Equation using a Graph

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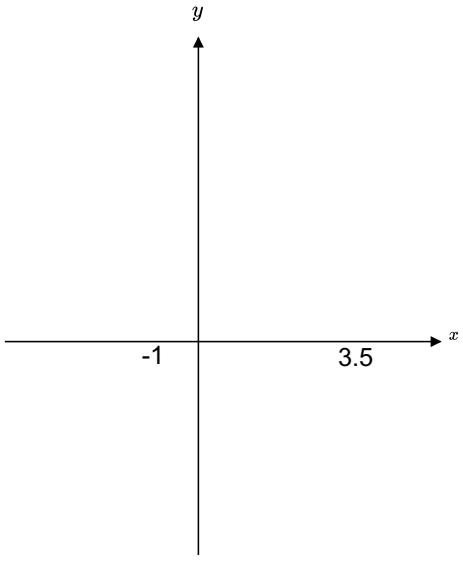


Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

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Further Example

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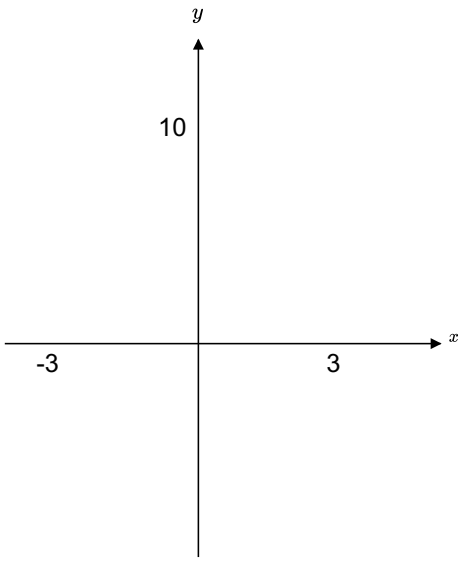
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

1.08 Graph Sketching: Quadratic Graphs

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Test Your Understanding

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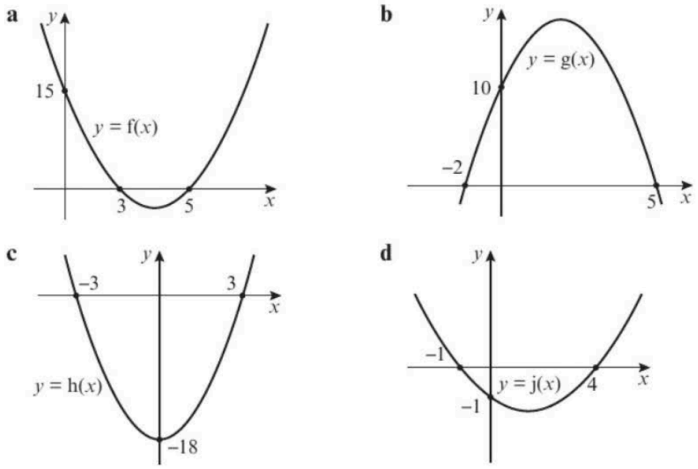
Determine an equation of this quadratic graph.

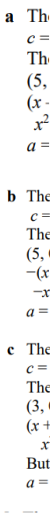
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2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a, b, c for each function.





1.08 Graph Sketching: Quadratic Graphs

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3 The graph of $y = ax^2 + bx + c$ has a minimum at $(5, -3)$ and passes through $(4, 0)$.
Find the values of a, b, c.

Extension Question:
[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?
$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

y

x